Question 3: Docker Storage - Demonstrate loss of container-layer data after container deletion, and permanent storage using a Docker named volume and bind mount.

Course outcome: CO1, CO2

Three storage demos were performed: (1) data written inside a container is lost when the container is deleted, (2) a named volume survives container deletion, (3) a bind mount syncs a host directory with the container.

## Demo 1 - container-layer data is lost

```
$ docker exec demo-layer sh -c "echo important-data > /tmp/note.txt && cat /tmp/note.txt"
important-data

$ docker rm -f demo-layer && docker run -d --name demo-layer ubuntu:24.04 sleep infinity
$ docker exec demo-layer cat /tmp/note.txt
cat: /tmp/note.txt: No such file or directory
--> file is GONE: container-layer data does not survive container deletion
```

## Demo 2 - named volume survives

```
$ docker volume create lab2data
$ docker run -d --name demo-vol -v lab2data:/data ubuntu:24.04 sleep infinity
$ docker exec demo-vol sh -c "echo persistent-data > /data/note.txt"
$ docker rm -f demo-vol && docker run -d --name demo-vol -v lab2data:/data ubuntu:24.04 sleep infinity
$ docker exec demo-vol cat /data/note.txt
persistent-data
--> file SURVIVED: the volume lives in /var/lib/docker/volumes/lab2data/_data
```

## Demo 3 - bind mount

```
$ echo "edited from the host" > ~/lab2-bindmount-data/host-note.txt
$ docker run --rm -v ~/lab2-bindmount-data:/data ubuntu:24.04 cat /data/host-note.txt
edited from the host

$ docker run --rm -v ~/lab2-bindmount-data:/data ubuntu:24.04 sh -c \
    "echo written-by-container >> /data/host-note.txt"
$ cat ~/lab2-bindmount-data/host-note.txt   (on the host)
edited from the host
written-by-container
--> both sides see the same file
```

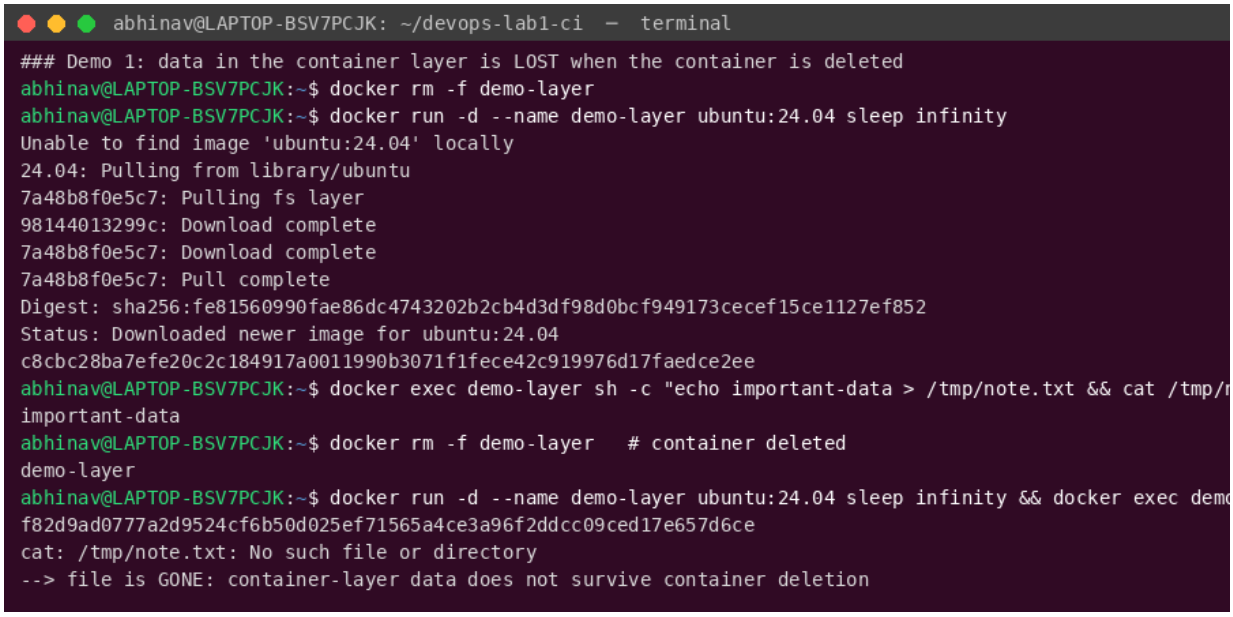



Fig: Demo 1: container-layer data lost after deletion

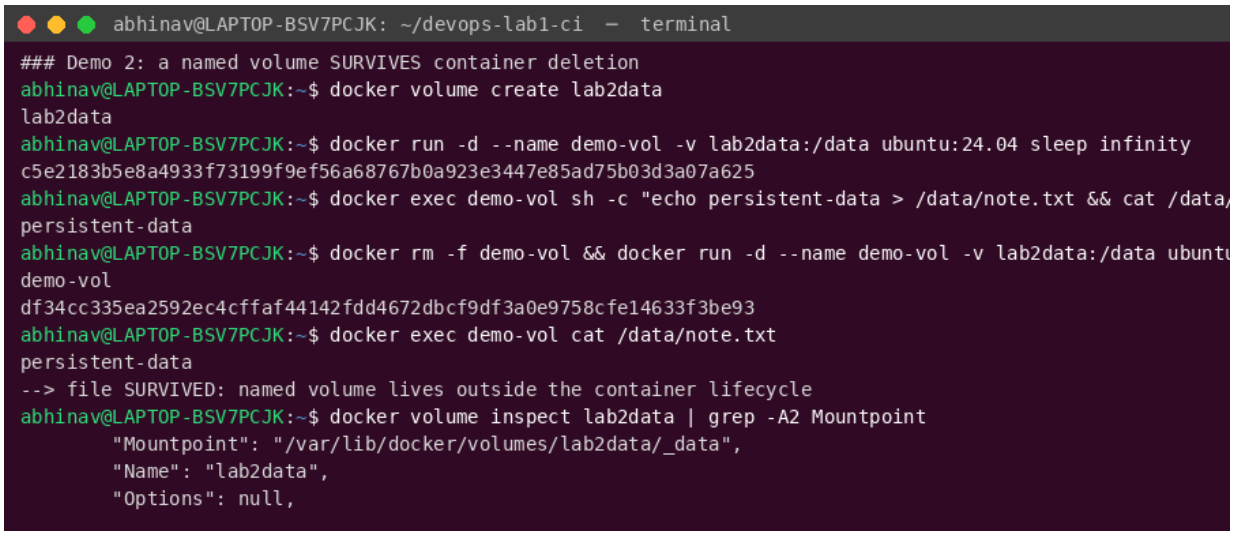



Fig: Demo 2: named volume survives container deletion

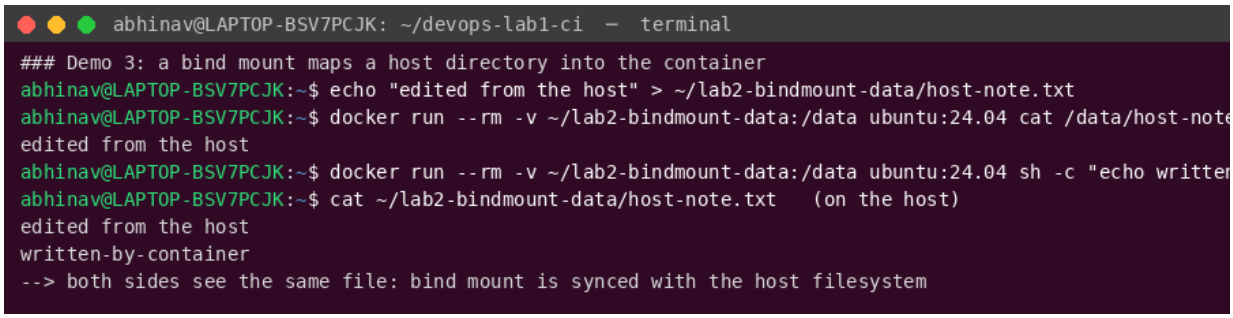



Fig: Demo 3: bind mount two-way sync

## Observations

- Writable container layers are ephemeral by design; anything that must survive belongs in a volume.
- Named volumes are managed by Docker (under /var/lib/docker/volumes/); bind mounts are plain host paths.
- Bind mounts are ideal for development (edit on host, see changes in container).

## Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Docker, Docker Compose, Minikube and kubectl. The evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-2 under the docs/ directory.